Distinct Colors

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \ldots, n$, and node 1 is the root. Each node has a color.

Your task is to determine for each node the number of distinct colors in the subtree of the node.

Input

The first input line contains an integer n: the number of nodes. The nodes are numbered $1, 2, \ldots, n$.

The next line consists of n integers $c_1, c_2, \ldots, c_n$: the color of each node.

Then there are $n-1$ lines describing the edges. Each line contains two integers a and b: there is an edge between nodes a and b.

Output

Print n integers: for each node $1, 2, \ldots, n$, the number of distinct colors.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le a, b \le n$
- $1 \le c_i \le 10^9$

Example

Input

```
5
2 3 2 2 1
1 2
1 3
3 4
3 5
```

Output

```
3 1 2 1 1
```